NCRTC Bus Management System

NCRTC Bus Management System — Initial System Design Questions

I. What is the primary objective of the application? Is it mainly for live bus tracking, daily operations management, incident handling, or managing all these activities through one platform?

II. Who are the intended users of the system? Which roles will use it, and what responsibilities will each role have?

III. What geographical and operational scope should the application cover? Will it manage multiple depots across NCR, and should users be able to access information across all depots or only their assigned depot?

IV. How should the four major modules interact with one another? For example, should an incident raised by a driver appear immediately in the control room, and should an incident linked to a bus connect to its trip history?

NCRTC Bus Management System — Initial Role-Specific Questions

A. Control-Room Operator

Primary responsibility: Monitor buses across all depots and coordinate responses to significant incidents.

1. What information should the operator access on the live dashboard? Should it include all buses, their current locations, speeds, assigned drivers, routes, and operational statuses?
2. Can the operator view buses across every depot, or should access be restricted to selected regions or operational areas?
3. What alerts should the operator receive? For example, should alerts include panic-button incidents, breakdowns, delayed buses, and vehicles outside their expected locations?
4. What actions can the operator perform when an incident occurs? Can they acknowledge an incident, assign it to a manager or another operator, change its status, and add notes?
5. What information should be available when the operator selects a bus on the map? Should it include the current driver, assigned route, recent location history, and related incidents?
6. Can the operator access historical trip information and incident records across all depots, and should they be able to download or export this information?

B. Depot Manager

Primary responsibility: Manage the buses, drivers, schedules, and incidents associated with a particular depot.

1. What information can the manager access about their depot? Should it include the fleet, drivers, routes, current bus locations, schedules, and open incidents?
2. Can a manager access information belonging to another depot, or is their visibility limited to their assigned depot?
3. How should the manager prepare and manage driver duties? Should they be able to assign a driver, vehicle, and route for a particular day or manage an entire weekly roster?
4. Can the manager modify published schedules, and what should happen if a driver or vehicle needs to be replaced after a duty has been published?
5. Which incidents can the manager handle? Can they view, assign, update, and resolve incidents only from their depot, including incidents raised by drivers?
6. What operational information should the manager see about drivers and vehicles? For example, should it include duty acknowledgements, vehicle availability, driver assignments, and unresolved incidents?

C. Driver

Primary responsibility: View assigned duties, acknowledge schedules, receive notices, and report incidents.

1. What information should the driver see after logging in? Should it include today's assigned depot, vehicle, route, reporting time, and trip details?
2. Can the driver view only their own assigned duties, or should they also be able to view future schedules and previous duty history?
3. What actions can the driver perform on an assigned duty? Can they acknowledge it, report an assignment problem, or request a change or leave?
4. How should a driver report an incident? Should they be able to select its type and severity, enter a description, attach a photograph, and associate the incident with their assigned vehicle?
5. What should happen when the driver activates the panic button? Should the system automatically create a critical incident and attach the driver's vehicle and available location information?
6. How should notices reach drivers? Should they be able to read notices targeted to all drivers or their depot, see unread notices, and acknowledge notices when required?

D. Administrator

Primary responsibility: Manage system-wide master data, user access, and official notices.

1. Which master records should the administrator be allowed to create, view, update, and delete? Should these include depots, vehicles, drivers, routes, stops, and user accounts?
2. Can the administrator manage operational records such as schedules and incidents, or should those responsibilities remain with depot managers and control-room operators?
3. How should user accounts and roles be managed? Can the administrator create accounts, deactivate users, change roles, and assign users to depots?
4. What information can the administrator access across the system? Should this include all depots, users, buses, schedules, trips, and incident records?
5. How should notices be managed? Can the administrator create, publish, schedule, edit, and withdraw notices, and target them to all drivers, a particular depot, or a particular role?
6. Should the administrator be able to track notice readership and acknowledgement, and inspect an audit history of important administrative or operational changes?

E. Executive / Read-Only User

Primary responsibility: Review summarized operational information for monitoring and decision-making.

1. Which reports and dashboard summaries should the executive be able to view? Should they include fleet activity, completed trips, incidents, and depot-level operational summaries?
2. Should the executive see information across all depots or only selected depots or regions?
3. What reporting period should be supported? For example, should reports cover the previous day, a selected date, a week, or a month?
4. What level of detail should be visible? Should the executive see aggregated statistics only, or also detailed incident records and individual bus-trip histories?
5. Should the executive be allowed to export or download reports, such as a PDF, while remaining unable to modify operational data?
6. Can the executive view live operational dashboards, or should access be limited to finalized historical reports?

NCRTC Bus Management System — Functional Requirements

Each requirement describes a specific capability the system must provide, rather than a technical implementation.

1. Driver

Primary responsibility: View assigned duties, acknowledge schedules, receive notices, and report incidents.

Functional requirement

I. View Assigned Duty: The system shall allow a driver to view their published duty for the current day, including the assigned depot, vehicle, route, and reporting time.

II. Acknowledge Duty: The system shall allow a driver to acknowledge an assigned duty and record the acknowledgement timestamp.

III. View Notices: The system shall display notices relevant to the driver, highlight unread notices, and record when a notice is read. It shall also support explicit acknowledgement when a notice requires it.

IV. Report Incidents: The system shall allow a driver to raise an incident by selecting its type and severity, entering a description, and optionally attaching a photograph and vehicle details.

V. Raise Emergency Alert: The system shall provide a panic button that automatically creates a P1 (critical) incident, associates the driver's assigned vehicle, and records the driver's location when available.

2. Depot Manager

Primary responsibility: Manage depot operations, driver duties, vehicles, and incidents.

Functional requirement

I. View Depot Operations: The system shall allow the depot manager to view the buses associated with their depot on a map, along with relevant vehicle details and available location information.

II. Manage Duty Rosters: The system shall allow the depot manager to view a weekly roster and assign a driver, vehicle, and route to a duty for a specified date.

III. Validate and Publish Duties: The system shall prevent a driver from being assigned to multiple duties on the same day and allow the manager to publish the roster so that assigned drivers can view their duties.

IV. Manage Incidents: The system shall allow the depot manager to view and filter incidents associated with their depot, inspect incident details, assign incidents to an authorized user, and update their status with a mandatory note for each status change.

V. View Bus and Trip History: The system shall allow the depot manager to select a vehicle and view its available location history for a specified date, including its recorded route on a map.

3. Administrator

Primary responsibility: Manage system-wide master data, user accounts, routes, and official notices.

I. Manage Master Data: The system shall allow the administrator to add, view, update, and delete vehicle and depot records, as well as manage driver and route information.

II. Manage User Accounts and Roles: The system shall allow the administrator to create and manage user accounts, assign roles, and associate users with their respective depots.

III. Manage Routes and Stops: The system shall allow the administrator to create, edit, and delete routes, including route names, associated depots, ordered stops, and planned arrival timings.

IV. Publish Notices: The system shall allow the administrator to create and publish notices with a title, message, publication date, and target audience, such as all drivers, a specific depot, or a specific role.

V. Track Notice Acknowledgements: The system shall allow the administrator to view which drivers have read or acknowledged a published notice, where acknowledgement is required.

4. Control-Room Operator

Primary responsibility: Monitor buses across all depots and coordinate responses to operational incidents.

I. Monitor Live Bus Locations: The system shall allow the control-room operator to view buses across all depots on a live map, with vehicle registration numbers, current speeds, and regularly updated locations.

II. View Bus Details and Location History: The system shall allow the operator to select a bus and view its assigned driver, current route, and recent GPS history. The operator shall also be able to view a vehicle's historical trip path for a selected date.

III. Monitor Operational Alerts: The system shall display relevant incidents and alerts, including emergency alerts raised through a driver's panic button, so the operator can identify situations requiring attention.

IV. Assign and Track Incidents: The system shall allow the operator to inspect incident details, acknowledge and assign serious incidents to authorized users, and track their progress through the incident workflow.

V. Update Incident Status: The system shall allow the operator to update an incident's status through the permitted workflow, add notes for status changes, and review its incident timeline

5. Executive (Read-Only)

Primary responsibility: Review operational reports and monitor overall bus operations without modifying system data.

1. View Daily Operational Report: The system shall allow the executive to view the previous day's operational report.

2. View Fleet Summary: The system shall allow the executive to view a summary of fleet operations, including available information about buses and their trips.

3. View Incident Summary: The system shall allow the executive to review summarized incident information to understand operational issues across the bus network.

4. Download Reports: The system shall allow the executive to download available operational reports in PDF format.

5. Read-Only Access: The system shall restrict the executive to viewing and downloading permitted reports without allowing them to create, modify, or delete operational records.

NCRTC Bus Management System — Non-Functional Requirements

A. Driver

Parameter	Non-functional requirement
Scalability	The system shall support the growing number of drivers and assigned duties while maintaining usability for a fleet of up to 500 vehicles.
Performance	The driver's duty and notices should load within 3 seconds, with normal API responses under 500 ms. The interface shall support mobile screens as narrow as 320 px.
Consistency (CP/AP)	CP-style consistency shall be prioritized for duty acknowledgements and incident submissions. Live location information may use AP-style behavior, displaying the last known location when updates are temporarily unavailable.
High Availability	The driver shall receive a clear indication when duties or notices cannot be fetched or submitted, without displaying a false success status.
Durability	Successfully recorded duty acknowledgements, incidents, and notice acknowledgements shall persist after application or server restarts.

B. Depot Manager

Parameter	Non-functional requirement
Scalability	The system shall support multiple depots and growing numbers of drivers, vehicles, routes, and duty assignments.
Performance	Weekly rosters and depot vehicle information should load within 3 seconds, with normal API responses under 500 ms.
Consistency (CP/AP)	CP-style consistency shall be prioritized for driver assignments, roster publication, and incident status changes to prevent conflicting operational records. Live map updates may favor availability.
High Availability	Roster management and incident handling should remain accessible during normal operations. Failed assignments or publication attempts must not be reported as successful.
Durability	Published rosters, driver assignments, and incident updates shall persist after server restarts, with recorded incident status changes retained in the history.

C. Control-Room Operator

Parameter	Non-functional requirement
Scalability	The system shall support monitoring all depots and up to 500 vehicles without making the live map unusable.
Performance	The live map shall refresh vehicle information every 5–10 seconds, as specified in the PRD. Normal API responses should remain under 500 ms.
Consistency (CP/AP)	AP-style behavior may be prioritized for live GPS visibility so the map remains available with clearly marked last-known positions. CP-style consistency shall be prioritized for incident assignments and status changes.

High Availability	Live monitoring and incident management should be highly available during bus operations. If location updates stop, the system should indicate stale data rather than present old positions as current.
Durability	Incident assignments, status transitions, notes, and available GPS history shall be persisted so that important operational events are not lost after a restart.

D. Administrator

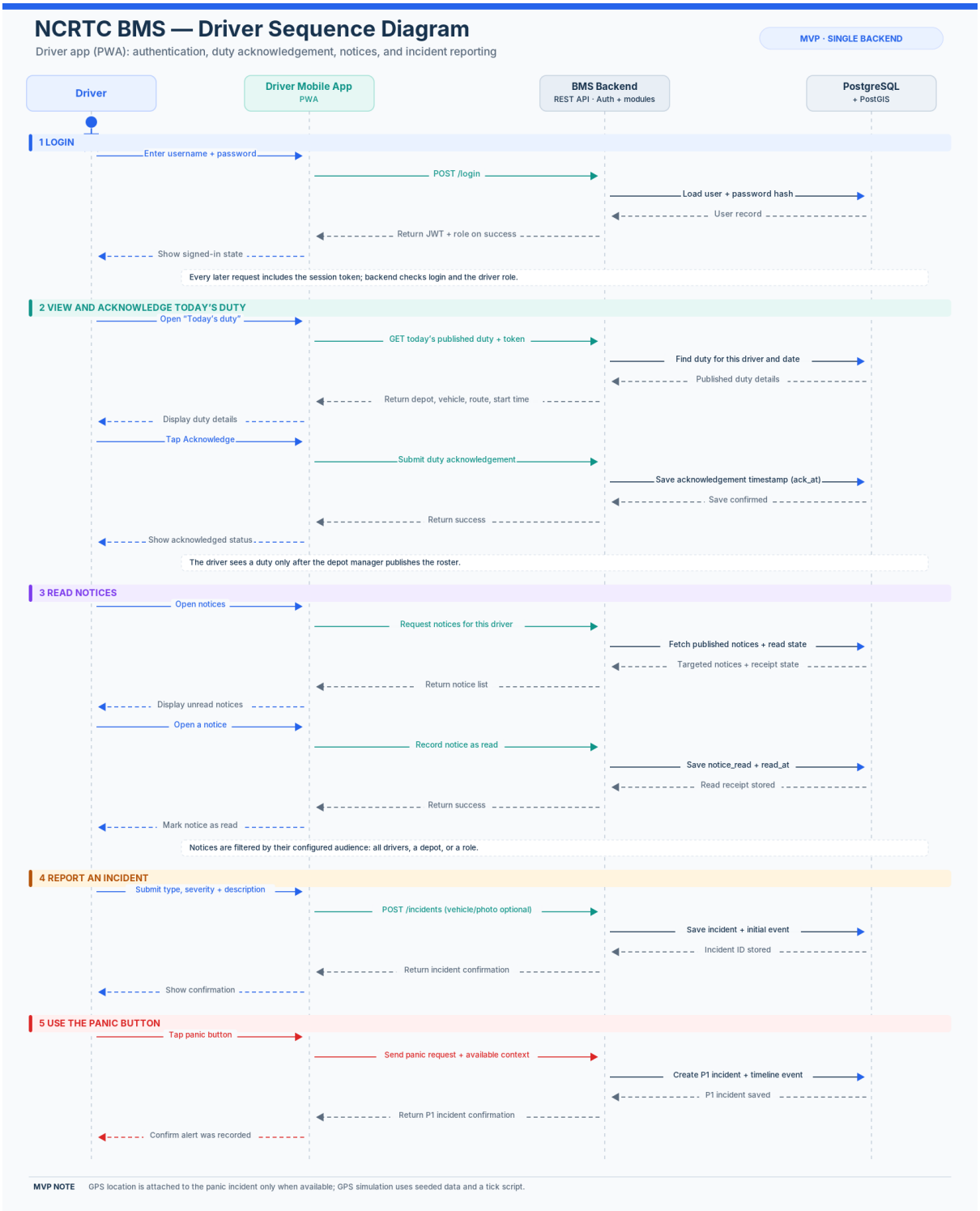
Parameter	**Non-functional requirement**
Scalability	The system shall accommodate increasing numbers of users, depots, routes, vehicles, and notices within the project's supported fleet size.
Performance	Master-data pages and user-management operations should load or respond within the common targets of 3 seconds for pages and 500 ms for normal API requests.
Consistency (CP/AP)	CP-style consistency shall be prioritized for user roles, access permissions, master-data changes, and notice publication to prevent conflicting or incorrect records.
High Availability	Administrative functions should be accessible during normal operating periods. Failed changes must return a clear error and must not leave partially updated records.
Durability	Successfully saved master data, user-account changes, and published notices shall remain stored after server restarts.

E. Executive (Read-Only)

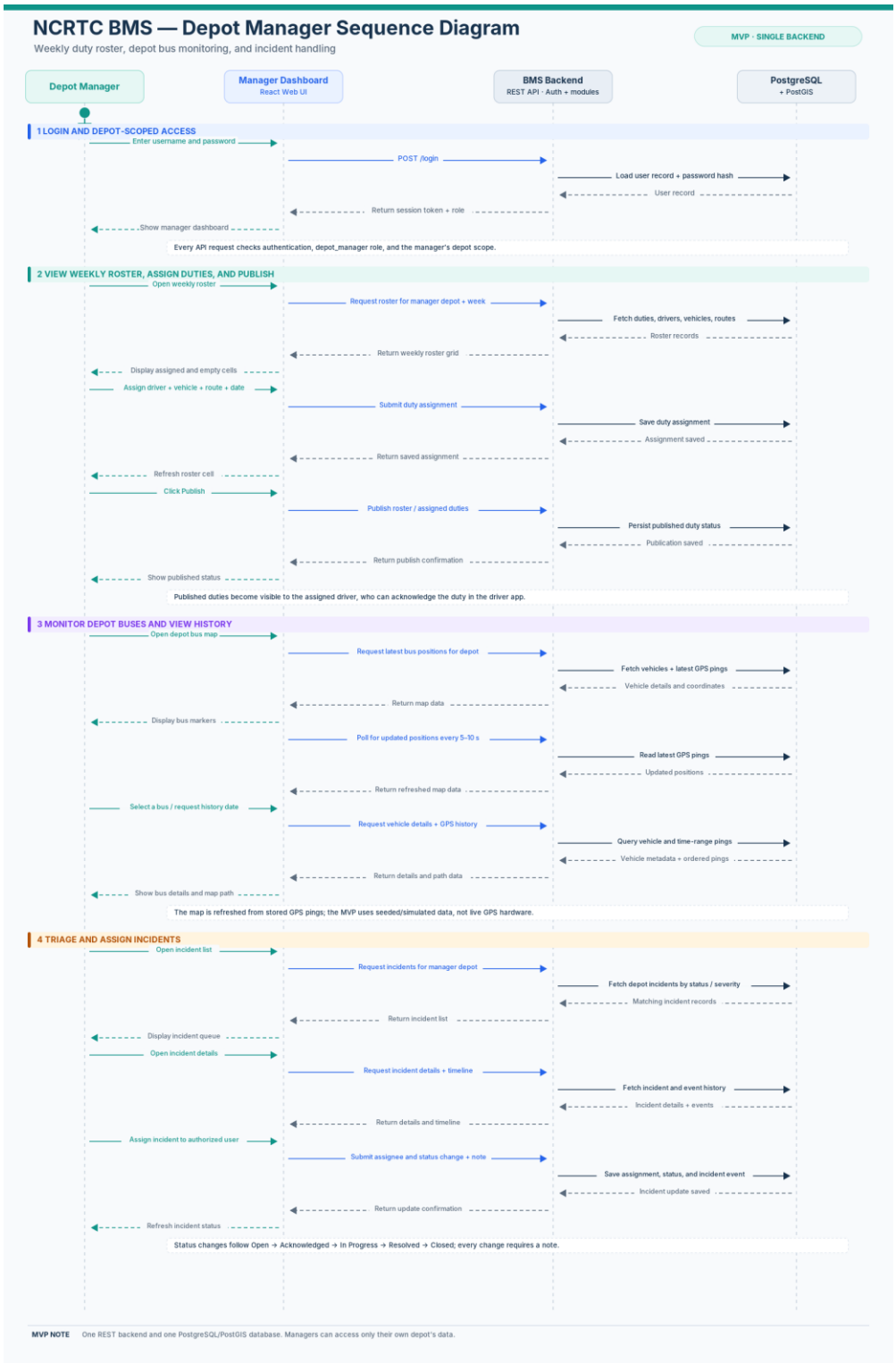
Parameter	**Non-functional requirement**
Scalability	The system shall generate operational summaries across multiple depots and a fleet of up to 500 vehicles.
Performance	Daily operational reports and dashboard summaries should load within 3 seconds, with normal API responses under 500 ms.
Consistency (CP/AP)	AP-style behavior may be acceptable for dashboard summaries, provided their last-refresh time is visible. Downloaded reports must use a consistent snapshot of the underlying data.
High Availability	Reports should be accessible during normal operating periods. Temporary reporting failures should produce a clear error rather than incomplete results presented as final.
Durability	Operational records used for reporting shall persist, and successfully generated report downloads shall contain the requested report rather than incomplete or corrupted output.

NCRTC Bus Management System — Sequence Diagrams

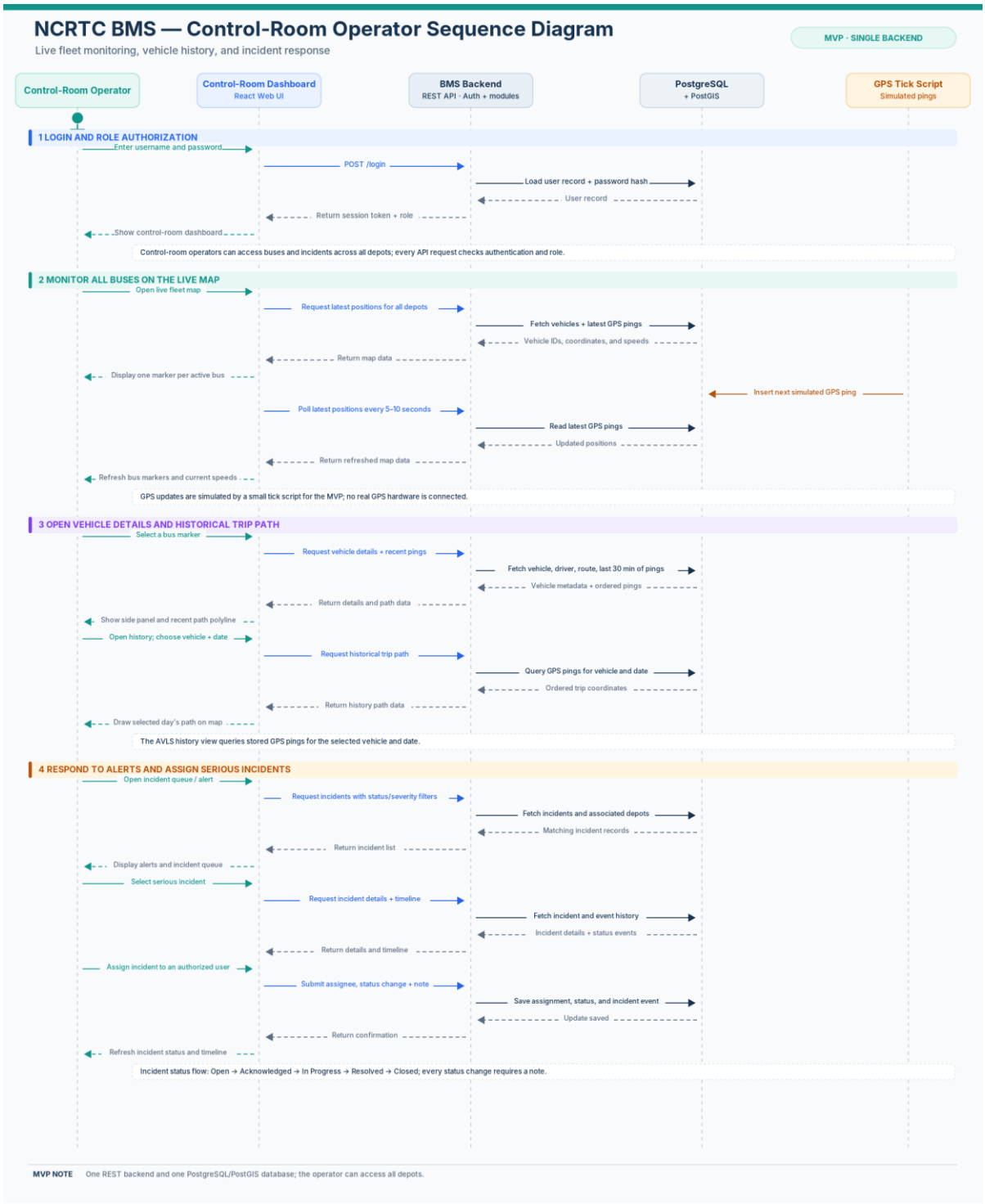
I. Driver



II. Depot Manager



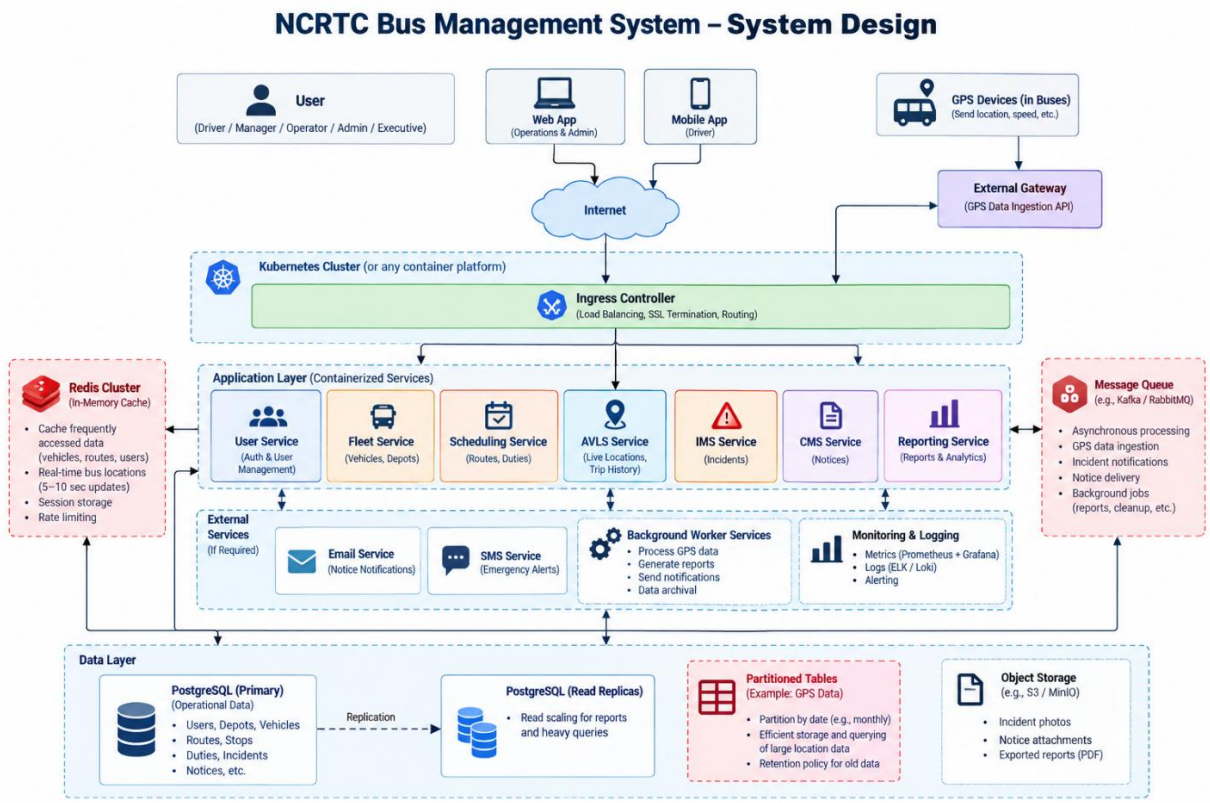
III. Control-Room Operator



IV. Admin


NCRTC BMS — Administrator Sequence Diagram
Master data, routes, user access, notices, and read receipts
MVP · SINGLE BACKEND
Administrator
Admin Dashboard
React Web UI
BMS Backend
REST API · Auth + modules
PostgreSQL
+ PostGIS
1 LOGIN AND ROLE AUTHORIZATION
Enter username and password
POST /login
Load user record + password hash
User record
Return session token + admin role
Show admin dashboard
Every protected API request checks authentication and role permissions before changing system data.
2 MANAGE MASTER DATA
Open vehicles, drivers, or depots
Request master-data list
Read selected records
Current records
Return list
Display records
Add or edit a record
Submit validated record details
Insert or update record
Save result
Return confirmation or validation error
Refresh list / show result
Master data includes depots, vehicles, and drivers, as scoped in the PRD.
3 CREATE AND MAINTAIN ROUTES AND STOPS
Open route management
Request routes and stops
Read route and ordered stop data
Routes, stops, planned timings
Return route data
Display route list and details
Create / edit / delete route
Submit route name, depot, ordered stops, timings
Save route and route-stop records
Save result
Return result
Refresh route view
Route fields include name, depot, ordered stops, and planned arrival times.
4 MANAGE USERS AND ROLES
Open user management
Request user list
Read users and role/depot associations
User records
Return user list
Display users
Create or update user and role
Submit user, role, and depot details
Save user account and role association
Save result
Return confirmation or validation error
Refresh user list
The PRD requires role-based access and depot associations; users must not gain access beyond their role.
5 PUBLISH NOTICES AND REVIEW READ RECEIPTS
Create notice: title, body, audience, publish date
Submit notice details
Save notice and target audience
Notice saved
Return publish confirmation
Show published notice
Open notice read-receipt list
Request readers / non-readers for notice
Join notice with notice_read records
Driver read receipts
Return read / not-read list
Display which drivers have read it
Notices may target all drivers, a specific depot, or a specific role. Read receipts record which drivers have or have not read a notice.
MVP NOTE
One REST backend and one PostgreSQL/PostGIS database; no separate admin service.

NCRTC Bus Management System — System Design



NCRTC Bus Management System — Sequence Diagrams

1. Authentication and user management

Method	Endpoint	Purpose	Access
POST	`/auth/login`	Authenticate a user and establish a JWT session	All users
GET	`/users`	List user accounts	Admin
POST	`/users`	Create a user account with a role and depot association	Admin
GET	`/users/{userId}`	View a user account	Admin
PATCH	`/users/{userId}`	Update permitted user-account details	Admin

2. Fleet and depot management

Method	Endpoint	Purpose	Access
GET	`/depots`	List depots within the caller's permitted scope	Admin, Control-Room Operator, Depot Manager, Driver
POST	`/depots`	Add a depot	Admin
GET	`/vehicles`	List buses, optionally filtered by depot	Admin, Control-Room Operator, Depot Manager
POST	`/vehicles`	Register a new vehicle	Admin
GET	`/vehicles/{vehicleId}`	Retrieve vehicle details	According to role and depot permissions

3. Route and scheduling management

Method	Endpoint	Purpose	Access
GET	`/routes`	List routes within the permitted scope	Admin, Depot Manager
POST	`/routes`	Create a route with ordered stops and planned timings	Admin
GET	`/routes/{routeId}`	View route details and its stops	Admin, Depot Manager
PATCH	`/routes/{routeId}`	Edit route details, stop order, or planned timings	Admin
DELETE	`/routes/{routeId}`	Delete a route	Admin
GET	`/stops`	Retrieve stops for route creation and display	Admin, Depot Manager
GET	`/duties/today`	View the logged-in driver's published duty for today	Driver
GET	`/duties?week_start={date}`	Retrieve the weekly duty roster	Depot Manager
POST	`/duties`	Assign a driver, vehicle, route, date, and duty timings	Depot Manager
PATCH	`/duties/{dutyId}`	Update a duty assignment	Depot Manager
POST	`/duties/{dutyId}/publish`	Publish a duty so the assigned driver can view it	Depot Manager

POST	`/duties/{dutyId}/acknowledge`	Record the driver's duty acknowledgement and timestamp	Driver

4. AVLS — Live tracking and trip history

Method	Endpoint	Purpose	Access
GET	`/vehicles/live-locations`	Retrieve the latest locations of active vehicles for the live map	Control-Room Operator, Depot Manager
GET	`/vehicles/{vehicleId}/live-details`	Retrieve registration number, current driver, route, and the last 30 minutes of GPS pings	Control-Room Operator, Depot Manager
GET	`/vehicles/{vehicleId}/location-history?date={date}`	Retrieve the vehicle's recorded path for a selected date	Control-Room Operator, Depot Manager
POST	`/internal/gps-pings`	Accept simulated GPS pings from the background tick script, if the script uses API-based ingestion	Internal simulation process only

The live-location endpoint can accept a depot_id filter for depot-specific views. The server must enforce the caller's depot permissions rather than trusting the filter supplied by the client.
The PRD calls for location updates every 5–10 seconds. This can initially be implemented through polling; a WebSocket endpoint is not required for the baseline project.

5. IMS — Incident Management

Method	Endpoint	Purpose	Access
GET	`/incidents`	List incidents, with filters for status, severity, depot, and `mine_only`	Authorized users
POST	`/incidents`	Raise an incident with type, severity, description, optional vehicle link, and optional photograph	Authenticated users
POST	`/incidents/panic`	Create a P1 incident automatically, associating the driver's current vehicle and location when available	Driver
GET	`/incidents/{incidentId}`	Retrieve incident details and its recorded status history	Authorized users
PATCH	`/incidents/{incidentId}/assignment`	Assign an incident to an authorized user	Depot Manager, Control-Room Operator
POST	`/incidents/{incidentId}/events`	Record an incident status transition and its mandatory note	Depot Manager, Control-Room Operator

The backend must enforce the incident workflow:

Open → Acknowledged → In Progress → Resolved → Closed

Every status change must include a note and create an incident-event record. Depot managers must remain within their depot's permitted scope, while control-room operators can handle incidents across depots. Drivers' incident access must respect their permitted data scope.

6. CMS — Notices and acknowledgements

Method	Endpoint	Purpose	Access
GET	`/notices`	Retrieve notices relevant to the logged-in user, optionally filtered to unread notices	Driver
GET	`/notices/{noticeId}`	View a notice	Driver, Admin
POST	`/notices`	Create and publish a notice with its title, body, target audience, and publication date	Admin
POST	`/notices/{noticeId}/read`	Record that a driver has read a notice and store the timestamp	Driver
POST	`/notices/{noticeId}/acknowledge`	Record explicit acknowledgement when required	Driver; optional feature
GET	`/notices/{noticeId}/reads`	Retrieve the read-receipt list for a notice	Admin

7. Reporting

Method	Endpoint	Purpose	Access
GET	`/reports/daily-operations`	Retrieve the previous day's operational report	Executive, Admin
GET	`/reports/daily-operations?format=pdf`	Download the operational report as a PDF	Executive, Admin

The PRD specifies yesterday's operational report and suggests PDF download. It does not define the report's exact metrics or detailed filtering capabilities.

8. Access-control rules for the entire API

These rules should apply across every module.

- Driver: Access assigned duties, relevant notices, incident reporting, and permitted depot-related information.
- Depot Manager: Access operational information and manage duties and incidents within their assigned depot.
- Control-Room Operator: Access fleet locations and operational incidents across depots, including assigning serious incidents.
- Admin: Manage users, vehicles, depots, routes, and notices, with system-wide access.
- Executive: Read-only access to operational reports.

The backend must verify authentication, role permissions, and depot scope on every protected request. Filtering data in the frontend alone is insufficient